



VCS VERIFICATION REPORT

SHANDONG PENGLAI PINGDINGSHAN WIND FARM PROJECT

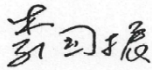


Document Prepared By

China Quality Certification Centre

Project Title	Shandong Penglai Pingdingshan Wind Farm Project
Version	2.0
Report ID	CDM-VCS-2022-0010

Report Title	VCS Verification Report of Shandong Penglai Pingdingshan Wind Farm Project
Client	Juno Carbon Investment & Environment Technology (Beijing) Co.,Ltd
Pages	49
Date of Issue	16-09-2022

Prepared By	China Quality Certification Centre
Contact	Address: No.9, No.188 South 4th Ring Road West, Fengtai District, Beijing,China Tel: +86 10 83886382 Email: zhengxianyu@cqc.com.cn Website: https://www.cqc.com.cn/
Approved By	Li Guozhen 
Work Carried Out By	Team Leader: Zheng Xianyu Team Member: Fang Qinglu/Zhou Wenshuo Technical Reviewer: Nie Xi

Summary:

China Quality Certification Centre (CQC) has performed the verification of the emission reductions reported for the project activity “Shandong Penglai Pingdingshan Wind Farm Project” (VCS Project ID: #258) for the monitoring period 26/10/2011-31/12/2016, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified 1 CL, 2 CARs and no FAR is raised in this monitoring period.

In CQC’s opinion, the GHG emission reductions reported for the project in the monitoring report (version 2.0 dated 28/08/2022) are fairly stated. The GHG emission reductions of the period from 26/10/2011-31/12/2016 were calculated correctly on the basis of approved methodology ACM0002 “Grid-connected electricity generation from renewable sources” (version 07), the monitoring plan contained in the registered VCS PD (i.e registered CDM PDD (version 03 dated 28/07/2006) for the 1st VCS crediting period 20/12/2008-20/07/2009 and the revised version(dated 24/02/2011).

CQC does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified.

The verification of reported emission reductions is based on the information made available to CQC and the engagement conditions detailed in this report. CQC cannot be held liable by any party for decisions made or not made based on this report.

Hence, CQC is able to certify that the emission reductions from the “Shandong Penglai Pingdingshan Wind Farm Project” during this monitoring period from 26/10/2011-31/12/2016 amount to 631,397 tCO₂e.

CONNETS

CONNETS	3
1 Introduction	5
1.1 Objective.....	5
1.2 Scope and Criteria	5
1.3 Level of Assurance.....	6
1.4 Summary Description of the Project	6
2 Verification Process.....	7
2.1 Method and Criteria	7
2.2 Document Review	7
2.3 Interviews	8
2.4 Site Inspections.....	9
2.5 Resolution of Findings	10
2.6 Eligibility for Validation Activities	11
3 Validation Findings.....	11
3.1 Participation under Other GHG Programs	11
3.2 Methodology Deviations.....	11
3.3 Project Description Deviations.....	11
3.4 Grouped Project	12
4 Verification Findings.....	13
4.1 Project Implementation Status	13
4.2 Safeguards	22
4.3 AFOLU-Specific Safeguards	23
4.4 Accuracy of GHG Emission Reduction and Removal Calculations	23
4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals	42
4.6 Non-Permanence Risk Analysis.....	42
5 Verification conclusion	42
APPENDIX A: REFERENCES	44

APPENDIX B: CORRECTIVE ACTION REQUESTS,CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS.....	47
---	-----------

1 INTRODUCTION

Juno Carbon Investment & Environment Technology (Beijing) Co.,Ltd. has commissioned China Quality Certification Centre (CQC) to carry out the verification and certification of emission reductions reported for the “Shandong Penglai Pingdingshan 48MW Wind Farm Project” (the Project) for the monitoring period 26/10/2011-31/12/2016. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Shandong Penglai Pingdingshan Wind Farm Project” for the period 26/10/2012-31/12/2016.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.1)^{/34/}
- VCS Standard (version 4.3)^{/33/} and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 07)^{/37/} applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to 'Project' as the expected annual emission reductions is 107,488 tCO₂e lower than 300,000 tCO₂e. Therefore, according to the requirement of para 4.1.8 (4) of VCS Standard Version 4.3 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CQC applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.1)^{/34/} and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: wind power project

Project Background

The Project is located in Beigou Town, Penglai City, Shandong Province, People's Republic of China. The geographical coordinates of the Project are east longitude of 120°38'48"~120°42'11" and north latitude of 37°44'09"~37°46'17". The nearest wind turbine to the substation is No.11, with east longitude of 120°42'01" and north latitude of 37°46'14". This project has been registered as a CDM project on 21/07/2009, and the registration number is #2397. The expected operation period of the Project is 20 years as stated in the registered PDD. CERs of 39,987 tCO₂e have been issued for the 1st monitoring period from 21/07/2009 to 24/11 /2009. CERs of 134,695 tCO₂e have been issued for the 2nd monitoring period from 25/11/2009 to 24/02/2011. CERs of 70,725 tCO₂e have been issued for the 3rd monitoring period from 25/02/2011 to 25/10/2011.

The Project involves the installation of 15 sets wind turbines with unit capacity of 2MW, and 10 sets wind turbines with unit capacity of 1.8MW, which totals up an installation capacity of 48MW, with a predicted power supplied to the grid of 101,904 MWh per year, and the annual full-load operation time amounts to 2,123h.

The purpose of the Project is to utilize a wind power facility to generate zero greenhouse gas (GHG) emissions electricity for the North China Power Grid (NCPG) thereby displacing electricity that is relatively carbon intensive. The project is therefore expected to reduce emissions of GHG by an estimated 107,488 tCO₂e per year during the first crediting period (20/12/2008-19/12/2018) by displacing electricity from the Grid.

The project has been registered as a VCS project with registration number of #258 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009

This VCS monitoring period covered the time from 26/10/2011 to 31/12/2016 and the emission reductions in this monitoring period are 631,397 tCO₂e.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (i.e. registered CDM PDD version 03 dated 28/07/2006) and monitoring plan contained in the registered VCS PD and the revised version, the applied methodology, and the VCS Standard (version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CQC appointed a verification team in accordance with CQC's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader	Zheng Xianyu	1.1, 1.2, 2.1	☒ DR ☒ SV ☒ RP ☐ TR
Team Member	Zhou Wenshuo	1.1, 1.2	☒ DR ☒ SV ☒ RP ☐ TR
	Fang Qinglu	1.1, 1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Nie Xi	1.2, 2.1, 13.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report^{/1/}, the registered VCS PD^{/13/}, emission reduction calculation spreadsheet^{/2/}, the following documents also were assessed as a part of the verification audit:

- The VCS validation report^{/24/};
- The previous VER/VCU monitoring report^{/25/} and verification report^{/26/}, and previous CDM/CER monitoring reports^{/27/} and verification reports^{/28/};
- Baseline and monitoring methodology ACM0002 (version 07) applied by the project^{/37/};
- Relevant decisions, clarifications and guidance from the Verra^{/33/-/46/}; and Other information and references relevant to the project activity.

During the desk review, CQC has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 08/08/2022, CQC visited the project proponent China Resources Wind Power (Yantai) Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team^{/37/}. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /38/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr.Gu Haiguang	Duty Director	China Resources Wind Power (Yantai) Co., Ltd.	Operation of the project activity;
Mr. ZhouWei	O&M Staff	China Resources Wind Power (Yantai) Co., Ltd.	Implementation of the monitor plan of the project activity;
Mr. Zhao Jie	Carbon asset manager	China Resources Wind Power (Yantai) Co., Ltd.	Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development

			contributions
Ms. Li Lin	Project Manager	Juno Carbon Investment & Environment Technology (Beijing) Co.,Ltd.	Data collection and ER calculation.
Ms. Ding Xue	Technical director	Juno Carbon Investment & Environment Technology (Beijing) Co.,Ltd.	

2.4 Site Inspections

During the on-site assessment, CQC has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, aggregating and reporting the monitoring parameters; and Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 07), and the management system were assessed during the site visit.

2.5 Resolution of Findings

The objective of this phase of the verification is to resolve issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the project activity to achieve emission removals or influence the monitoring and reporting of emission removals prior to CQC's positive conclusion on the GHG emission removals calculation.

Findings established during the verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

A Corrective Action Request (CAR) is raised, if one of the following situations occurs:

- (a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;
- (b) Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants;
- (c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- (d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

A Clarification Request (CL) is raised, if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A Forward Action Request (FAR) is raised, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

To guarantee the transparency of the verification process, the concerns raised are documented in more detail in Appendix A.

The verification team identified 1 CL and 2 CARs in this monitoring period. The CL and CAR were satisfactorily addressed by the project proponents in the updated monitoring report (version 2.0 dated 28/08/2022) (refer to Appendix A for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CQC confirmed that there was no FAR identified in previous verification^{/26/}, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by TÜV SÜD Industrie Service GmbH under the CDM rules and registered as CDM project on 21/07/2009, and was registered as VCS project under VCS standard version 2007.1. It has three monitoring periods during the first CDM crediting period. The emission reduction of the first all monitoring periods has been issued. By checking the UNFCCC website and the previous verification reports^{/26/}, it is confirmed that these monitoring periods are all before this monitoring period (26/10/2011-31/12/2016) and the verifications are not done by the CQC.

CQC was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the proposed project. The verification for the monitoring period 26/10/2011-31/12/2016 has been completed by CQC. In combination of this monitoring period 26/10/2011-31/12/2016, the verification undertaken by CQC for the total reporting period 26/10/2016-30/11/2021 is no more than consecutive 6 years., thus in compliance with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.3.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a Clean Development Mechanism (CDM) project in UNFCCC on 21/07/2009 (UNFCCC Ref. 2397), with the first 7 years crediting period started from 21/07/2009. CERs of 39,987 tCO₂e have been issued for the 1st monitoring period from 21/07/2009 to 24/11/2009. CERs of 134,695 tCO₂e have been issued for the 2nd monitoring period from 25/11/2009 to 24/02/2011. CERs of 70,725 tCO₂e have been issued for the 3rd monitoring period from 25/02/2011 to 25/10/2011.

The project has been registered as a VCS project with Ref. VCS 258 on 20/12/2008 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009.

All credits from 26/10/2011 to 31/12/2016 will be claimed under VCS program as VCUs for the project to avoid double counting.

3.2 Methodology Deviations

The validation process^{/17/} has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 07)^{/37/}. There was not any methodology deviation applied to this project.

3.3 Project Description Deviations

In the registered VCS PD, the crediting period is described as from 20/12/2008 to 20/07/2009. A deviation is requested for the crediting period in the registered VCS PD. The project is registered under VCS Standard 2007 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS 2007 which shall be a maximum of ten years

and may be renewed at most twice, so the first renewable crediting period of the project should be updated from 20/12/2008 - 20/07/2009 to 20/12/2008-19/12/2018, and it has no impacts on the applicability, additionality or baseline of the project.

After the first CDM monitoring period (from 21/07/2009 to 24/11/2009) of the Project, it was noticed that the Project shared the same 220kV substation (Tangqiu 220kV Substation) with the other wind farm, namely Shandong Penglai Wind Farm Phase II Project (hereafter refers to Project B), which was put into operation on 25/11/2009. The main gateway meter (M1) installed at the Tangqiu 220kV Substation monitors the electricity supplied to and imported from the grid by the Project and Project B. Separate meters (M4, M5, M6, M7 and M8) have been installed at each project site to measure electricity supplied. The grid company calculates the net electricity supplied by the Project based on the meter readings of the gateway meter (M1) and these on-site separate meters (M7, M8) of Project B.

Considering the actual monitored parameters have not been specified in the monitoring plan in the registered PDD. The monitoring plan of the Project has been revised to including actual monitoring parameters. The revised monitoring plan was approved by EB on 24/02/2011.

There is no other project description deviation in this monitoring period. CQC confirms that the project is fully implemented in accordance with the project description, and no any project description deviations. Details refer to section 1.16.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Shandong Penglai Pingdingshan Wind Farm Project” for the monitoring period 26/10/2011-31/12/2016.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

Shandong Penglai Pingdingshan Wind Farm Project (hereafter refers to “the project”) developed by China Resources Wind Power (Yantai) Co., Ltd. is a grid-connected renewable energy Project in Beigou Town, Penglai City, Shandong Province, P.R. China.

The Project involves the installation of 15 sets wind turbines with unit capacity of 2MW, and 10 sets wind turbines with unit capacity of 1.8MW, which totals up an installation capacity of 48MW, with a predicted power supplied to the grid of 101,904 MWh per year, and the annual full-load operation time amount to 2,123h.

The purpose of the Project is to utilize a wind power facility to generate zero greenhouse gas (GHG) emissions electricity for the North China Power Grid (NCPG) thereby displacing electricity that is relatively carbon intensive. The project is therefore expected to reduce emissions of GHG by an estimated 107,488 tCO₂e per year during the first crediting period (20/12/2008-19/12/2018) by displacing electricity from the Grid.

The scenario existing prior to the start of the implementation of the project is: The same electricity output by the project activity would have otherwise been generated by the operation of NCPG connected power plants and by the addition of new generation sources. That is the same as the baseline scenario.

The Project started to construct on 25/04/2008. It started commissioning on 20/12/2008, and was put into full delivery on 21/01/2009.

The project has been registered as a CDM project on 21/07/2009, and the reference number is #2397. CERs of 245,407 tCO₂e have been issued for the monitoring period from 21/07/2009 to 25/10/2011.

The project has been registered as a VCS project with VCS ID of #258 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009. This VCS monitoring period covered the time from 26/10/2011 to 31/12/2016 and the emission reductions in this monitoring period are 631,397 tCO₂e.

On-site training for the related procedures including, recording and reporting was verified to be in place^{/6/} and their implementation was confirmed by interview with the key operators^{/38/} and observing the operation.

CQC confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is

confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

Compliance of monitoring plan with monitoring methodology

CQC is able to confirm that the monitoring plan contained in the VCS PD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 07).

Compliance of monitoring with the monitoring plan

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD, CQC confirms that all parameters stated in the monitoring plan are monitored and reported appropriately. All parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 07) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex ante” in the MR are checked against the registered VCS PD. The parameter $EF_{grid,CM,y}$ is the combined emission factor of the grid, which was determined ex ante at the validation stage and has been fixed during the first crediting period. CQC verified and confirms that the emission factor used in the monitoring report is in compliance with the VCS PD.

Parameters monitored

CQC conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and electricity receipts;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan, the parameters to be monitored is as below:

$EG_{export,project}$	Electricity supplied by the project activity to the grid
$EG_{import,project}$	Electricity imported from the grid by the project
$EG_{export,total}$	Electricity supplied by the Project and Project B to the grid

$EG_{import,total}$	Electricity imported from the grid to the Project and Project B
$EG_{export,other,y}$	Electricity supplied by the Project B to the grid
EG_y	Net electricity supplied to the grid by the project activity

Monitored parameters $EG_{export,project}$ and $EG_{import,project}$

By site inspection and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CQC confirmed $EG_{export,project}$ and $EG_{import,project}$ are continuously measured with electricity meters M4, M5, and M6 installed on the collection line 1, 2, and 3. The accuracy of above meters is 0.2S. The records of the meters are based on continuous measurement and monthly recorded by the project owner. The meter readings of electricity exports and imports were continuously measured, daily recorded and monthly aggregated.

The verification team had on-site checked the location of the meters against the diagram of power connection system and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CQC during on-site visit.

By checking operation log and via on-site visit, it is confirmed that all meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records, monthly and daily reading records and electricity receipts issued by grid company, it is confirmed by the verification team that the electricity is continuously measured by M4, M5 and M6. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th. The electricity receipts issued by the grid company have been used for verifying the records of $EG_{export,project}$ and $EG_{import,project}$ measured by electricity meters M4, M5, and M6.

Monitored parameter $EG_{export,total}$ and $EG_{import,total}$

By site interview and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CQC confirmed $EG_{export,total}$ and $EG_{import,total}$ are continuously measured by main meter M1 and backup meter M2. Both meters are bi-directional and installed at Tangqiu 220kV Substation. The meter reading was daily recorded, monthly aggregated and archived electronically. By checking operation log monthly and daily reading records and via on-site visit, it is confirmed that in this monitoring period no electricity was consumed by the project through the backup line.

The verification team had on-site checked the location of the meters against the diagram of power connection system^{14/} and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CQC during on-site visit.

By checking operation log^{7/} and via on-site visit, it is confirmed that both meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records^{7/}, monthly and daily reading records and electricity receipts issued by grid company, it is confirmed by the verification team that the electricity is continuously measured by main meter M1 and backup meter M2. The cut-off time of recording of all meters is at 24:00 on the

last day of each month, except from November 2011 to August 2012, which is at 24:00 on the 24th or 25th/8/. The electricity receipts issued by the grid company have been used for verifying the records of $EG_{\text{export,total}}$ and $EG_{\text{import,total}}$ measured by electricity meters M1 and M2.

Monitored parameter $EG_{\text{export,other,y}}$

By site interview and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CQC confirmed $EG_{\text{export,other,y}}$ is continuously measured by main meter M7 and M8, and backup meter M9 and M10. All meters are bi-directional and installed on the collection lines 4 and 5. The accuracy of the main meters (M7 and M8) and backup meters (M9 and M10) is all 0.5s. The meter reading was daily recorded, monthly aggregated and archived electronically. By checking operation log^{7/} monthly and daily reading records/8/ and via on-site visit, it is confirmed that in this monitoring period no electricity was consumed by the project through the backup line.

The verification team had on-site checked the location of the meters against the diagram of power connection system and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CQC during on-site visit. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th.

As described above, the meters have been installed in accordance with VCS Program rules and meet all appropriate rules and requirements of VCS standard. CQC has on-site checked the location of the meters against the diagram of power connection system and found them to be consistent.

Data in the monthly reading records were used to the report, through a cross check with Electricity Receipts. The conservative values were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

CAR 01 was raised because of the monitoring period in the MR (from 25/06/2012 to 19/12/2018, which is more than six consecutive years) does not meet the requirements of rotation of validation/verification bodies in VCS-Standard_v4.3. The monitoring period has revised from 25/06/2012-19/12/2018 to 26/10/2011-31/12/2016, thus **CAR 01** was subsequently successfully closed. Refer to Appendix B for details.

CAR 02 was raised because of information of the monitoring meters installed during the monitoring period is not provided in the MR and supporting documents. The information of the monitoring meters installed during the monitoring period has been added in Section 4.2 of the revised MR, and the supporting document has been replenished, thus **CAR 02** was subsequently successfully closed. Refer to Appendix B for details.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 07) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Meters	Type	SN	Accuracy class	Calibration Date	Validation	Calibration Entity
M1	ZMQ202 C	94089552	0.2S	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center
M2	DTSD34 1	200807780 20034	0.5S	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
M4	DTSD34 1	090901672 90077	0.2S	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center
M5	DTSD34 1	090901672 90078	0.2S	01/09/2011; 11/09/2012;	From 2011 to 2014, it was verified by Yantai Power Supply Company	

				10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
M6	DTSD34 1	090901672 90079	0.2S	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
M7	DTSD34 1	200908326 835	0.5S	09/11/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
M8	DTSD34 1	200908326 832	0.5S	09/11/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power

					Power Center	Metering Center	Metering Center
M9	DTSD34 1	200908326 874	0.5S	09/11/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2015	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center		
M10	DTSD34 1	200908326 873	0.5S	09/11/2011, 11/09/2012, 10/09/2013, 11/03/2014, 10/03/2015; 08/03/2016	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center		

Calibration records and accreditation certificates^{/14//15/} have been verified by the verification team. The actual accuracy of meters M1, M4, M5 and M6 is verified to be 0.2S, the actual accuracy of meters M2, M7, M8, M9 and M10 is verified to be 0.5S, which is consistent with the requirement in the monitoring plan contained in the registered VCS PDD and in compliance with the Technical administrative code of electric energy metering^{/16/}. The calibration frequency of meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CQC confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan. And the calibrations are verified to be valid for the whole reporting period.

Data management and control

China Resources Wind Power (Yantai) Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control

procedures have been addressed in the project management and monitoring manual, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/38/ and check records/7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the VCS PD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

By checking the UNFCCC website (<https://cdm.unfccc.int/Projects/DB/RWTUV1234176728.19/view>), it is confirmed that the project has been registered as a Clean Development Mechanism (CDM) project in UNFCCC on 21/07/2009 (UNFCCC Ref. 2397), with the first 7 years crediting period started from 21/07/2009. CERs of 39,987 tCO₂e have been issued for the 1st monitoring period from 21/07/2009 to 24/11/2009. CERs of 134,695 tCO₂e have been issued for the 2nd monitoring period from 25/11/2009 to 24/02/2011. CERs of 70,725 tCO₂e have been issued for the 3rd monitoring period from 25/02/2011 to 25/10/2011. The project has been registered as a VCS project with Ref. VCS 258 on 20/12/2008 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 26/10/2011-31/12/2016 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCUs issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction(CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tons of CO₂e/year as per the regulation^{/18/} including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license^{/15/} of the project proponent, it is confirmed that the PP is in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no mission cap enforced for the PP by further checking the enforced

company list under China ETS in public information^{/21/}. Moreover, it is confirmed by CQC the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/18/. Hence, it is confirmed that the emission reductions will not be double counted.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption.
The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the grid NCPG. During this monitoring period, 598,594.662 MWh of net electricity generated by wind energy was delivered to the NCPG. It is verified by checking the daily and monthly reading records^{/8/} and via on-site inspection. Meanwhile, by checking the approved VCS and CDM monitoring reports for the previous monitoring, the project has supplied 666,216.962 MWh renewable electricity to NCPG during the first VCS monitoring period (20/12/2008~20/07/2009) and this monitoring period. The project is expected to supply 2,038,080 MWh renewable energy to NCPG during the operation lifetime which contribute the SDGs 7. This contributes to increasing the share of clean energy consumption of China's National Plan on Implementation of 2030 Agenda for Sustainable Development (<http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>).
- SDG 8, Indicator 8.3.1 Proportion of informal employment in non agriculture employment, by sex.
It is verified by interviewing with project proponent and staff and checking the staff roster^{/22/}, 24 people including 5 females were employed by the project owner for the operation, monitoring and equipment maintenance of this project during this monitoring period. The project activity had provided temporary working opportunities during the construction period. From the operation start date, 24 people were employed including 5 females during the first VCS monitoring period and this monitoring period. This contributes to "Increase labor force participation rate through implementation rate through implementation of the classification policy. Vigorously enforce the law on promotion of employment" as stated in China's National Plan on Implementation of 2030 Agenda for Sustainable Development (<http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>).
- SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.
As verified by on-site inspection, and via checking the emission reduction calculation sheet^{/2/} and the daily and monthly reading records^{/8/}, 631,397tCO₂e GHG emission reductions are generated by the project activity during this monitoring period. And by checking the VCS and CDM monitoring and verification reports for the previous monitoring periods^{/27//28/}, it is confirmed the cumulative emission reductions are 702,725 tCO₂e during the first VCS monitoring period and this monitoring period. And according to the registered PDD, the project was expected to prevent the release of 2,149,760 tCO₂e into the atmosphere during the project operation lifetime of 20 years. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas".

In conclusion, Verification Team was able to confirm that the project implementation is in

accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, wastewater, noise, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent and staff and checking the staff roster^[22], that the project has contributed to the local employment by providing job opportunities. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training organised by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

4.2.2 Local Stakeholder Consultation

LSC prior to the project implementation

On 17 February 2008, the project owner carried out a survey of the local residents around the project location. The consultation invited local stakeholders to submit comments on the Project activity by filling in a questionnaire sent out by the Project Developer. The staff introduced the background of the proposed project and then sent out 50 copies of questionnaire in a random way, 50 pieces of reply were received. Among the interviewees, 31 of them are farmers, 10 are workers, 8 are governmental officials and 1 student.

An invitation notice for stakeholder comments was later issued by the project developer on March 1st, 2008 and 19 representatives of local stakeholders, including governmental officials of local County and local residents, etc. attended the meeting on March 11 to discuss the questionnaires collected and further introduce the project. No negative opinion on construction of the project is heard and environmental considerations expressed by stakeholders are discussed on the meeting.

The stakeholder meeting discussed the questionnaires received, from which it shows the local community possesses basically positive comments on the effects of the proposed project. They believe that there will be improvement for local economic development. At the same time, they also express consideration about potential impact on environment. As to noise, the distance between a local plant to one of the original wind turbine locations is very near, considering the noise interruption in the future, the turbine has been moved to other place, thus to mitigate the noise.

LSC during the operation period

To keep on-going communications with local stakeholders, the project owner has public its phone number of the central control room at the power plant gate and also oral noticed to local people and village committees. Anyone who have comments on the project could phone the project owner directly. Besides, the project owner also put a grievance book in the office of wind plant and the villages can leave their grievance or suggestions on the book anytime.

Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations.

There are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CQC confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

According to ACM0002 and the registered VCS PD of the Project, the baseline emission BE_y during the monitoring period results from:

$$BE_y = EG_y * EF_{grid,CM,y}$$

Where:

EG_y – Net electricity supplied to the grid by the project activity in year y;

$EF_{grid,CM,y}$ - Combined margin CO_2 emission factor for the project electricity system in year y (tCO_2e/MWh).

The verification team has checked the daily and monthly reading records, electricity receipts, Confirmation letter for electricity readings issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and EG_y calculation are summarized in Table 1 as follow :

Table 1: Electricity supplied by the Project and Project B to the grid and Electricity imported from the grid to the Project and Project B

Monitoring period	EG _{export,total}			EG _{import,total}		
	Data based on meter readings M1	Data based on sales receipt	Conservative value	Data based on meter readings M1	Data based on sales receipt	Conservative value
	MWh	MWh	MWh	MWh	MWh	MWh
	A	B	C	D	E	F
26/10/2011-24/11/2011	20,193.200	20,193.200	20,193.200	60.720	60.720	60.720
25/11/2011-25/12/2011	20,971.160	20,971.160	20,971.160	52.800	52.800	52.800
26/12/2011-31/12/2011	3,405.600	3,405.600	3,405.600	0.000	0.000	0.000
Subtotal 2011	-	-	44,569.960	-	-	113.520
01/01/2012-25/01/2012	9,643.920	9,643.920	9,643.920	0.000	0.000	0.000
25/01/2012-24/02/2012	18,831.120	18,831.120	18,831.120	150.480	150.480	150.480
25/02/2012-25/03/2012	18,987.800	18,987.800	18,987.800	47.520	47.520	47.520
26/03/2012-24/04/2012	32,244.960	32,244.960	32,244.960	26.400	26.400	26.400
25/04/2012-25/05/2012	19,272.000	19,272.000	19,272.000	71.280	71.280	71.280
24/05/2012-30/06/2012	25,098.480	25,098.480	25,098.480	63.360	63.360	63.360
01/07/2012-31/07/2012	14,688.960	14,688.960	14,688.960	84.480	84.480	84.480
01/08/2012-23/08/2012	6,473.280	6,473.280	6,473.280	105.600	105.600	105.600
24/08/2012-30/09/2012	18,004.800	18,004.800	18,004.800	92.400	92.400	92.400
01/10/2012-31/10/2012	19,715.520	19,715.520	19,715.520	52.800	52.800	52.800
01/11/2012-30/11/2012	26,967.600	26,967.600	26,967.600	34.320	34.320	34.320
01/12/2012-31/12/2012	24,546.720	24,546.720	24,546.720	39.600	39.600	39.600
Subtotal 2012	-	-	234,475.160	-	-	768.240
01/01/2013-31/01/2013	18,308.400	18,308.400	18,308.400	42.240	42.240	42.240

01/02/2013-28/02/2013	12,608.640	12,608.640	12,608.640	147.840	147.840	147.840
01/03/2013-31/03/2013	31,658.880	31,658.880	31,658.880	18.480	18.480	18.480
01/04/2013-30/04/2013	30,101.280	30,101.280	30,101.280	29.040	29.040	29.040
01/05/2013-31/05/2013	23,617.440	23,617.440	23,617.440	34.320	34.320	34.320
01/06/2013-30/06/2013	19,430.400	19,430.400	19,430.400	58.080	58.080	58.080
01/07/2013-31/07/2013	13,453.440	13,453.440	13,453.440	81.840	81.840	81.840
01/08/2013-31/08/2013	14,343.120	14,343.120	14,343.120	76.560	76.560	76.560
01/09/2013-30/09/2013	11,993.520	11,993.520	11,993.520	100.320	100.320	100.320
01/10/2013-31/10/2013	20,787.360	20,787.360	20,787.360	42.240	42.240	42.240
01/11/2013-30/11/2013	22,672.320	22,672.320	22,672.320	58.080	58.080	58.080
01/12/2013-31/12/2013	24,192.960	24,192.960	24,192.960	18.480	18.480	18.480
Subtotal 2013	-	-	243,167.760	-	-	707.520
01/01/2014-31/01/2014	20,132.640	20,132.640	20,132.640	23.760	23.760	23.760
01/02/2014-28/02/2014	21,209.760	21,209.760	21,209.760	52.800	52.800	52.800
01/03/2014-31/03/2014	21,492.240	21,492.240	21,492.240	34.320	34.320	34.320
01/04/2014-30/04/2014	18,973.680	18,973.680	18,973.680	58.080	58.080	58.080
01/05/2014-31/05/2014	29,924.400	29,924.400	29,924.400	29.040	29.040	29.040
01/06/2014-30/06/2014	11,528.880	11,528.880	11,528.880	60.720	60.720	60.720
01/07/2014-31/07/2014	15,253.920	15,253.920	15,253.920	81.840	81.840	81.840
01/08/2014-31/08/2014	6,378.240	6,378.240	6,378.240	132.000	132.000	132.000
01/09/2014-30/09/2014	9,065.760	9,065.760	9,065.760	81.840	81.840	81.840
01/10/2014-31/10/2014	22,297.440	22,297.440	22,297.440	52.800	52.800	52.800
01/11/2014-30/11/2014	15,768.720	15,768.720	15,768.720	39.600	39.600	39.600
01/12/2014-31/12/2014	29,480.880	29,480.880	29,480.880	44.880	44.880	44.880
Subtotal 2014	-	-	221,506.560	-	-	691.680

01/01/2015-31/01/2015	23,810.160	23,810.160	23,810.160	18.480	18.480	18.480
01/02/2015-28/02/2015	16,101.360	16,101.360	16,101.360	60.720	60.720	60.720
01/03/2015-31/03/2015	25,130.160	25,130.160	25,130.160	42.240	42.240	42.240
01/04/2015-30/04/2015	28,511.470	28,511.470	28,511.470	39.600	39.600	39.600
01/05/2015-31/05/2015	22,968.000	22,968.000	22,968.000	54.120	54.120	54.120
01/06/2015-30/06/2015	16,518.220	16,518.220	16,518.220	80.784	80.784	80.784
01/07/2015-31/07/2015	11,064.240	11,064.240	11,064.240	100.056	100.056	100.056
01/08/2015-31/08/2015	5,733.290	5,733.290	5,733.290	120.120	120.120	120.120
01/09/2015-30/09/2015	7,661.280	7,661.280	7,661.280	135.696	135.696	135.696
01/10/2015-31/10/2015	18,996.910	18,996.910	18,996.910	69.168	69.168	69.168
01/11/2015-30/11/2015	22,885.900	22,885.900	22,885.900	56.496	56.496	56.496
01/12/2015-31/12/2015	21,495.140	21,495.140	21,495.140	53.592	53.592	53.592
Subtotal 2015	-	-	220,876.130	-	-	831.072
01/01/2016-31/01/2016	22,600.780	22,600.780	22,600.780	68.904	68.904	68.904
01/02/2016-28/02/2016	15,923.690	15,923.690	15,923.690	43.560	43.560	43.560
01/03/2016-31/03/2016	22,322.520	22,322.520	22,322.520	34.320	34.320	34.320
01/04/2016-30/04/2016	24,050.660	24,050.660	24,050.660	27.720	27.720	27.720
01/05/2016-31/05/2016	18,280.680	18,280.680	18,280.680	27.456	27.456	27.456
01/06/2016-30/06/2016	19,477.390	19,477.390	19,477.390	77.352	77.352	77.352
01/07/2016-31/07/2016	14,986.750	14,986.750	14,986.750	77.352	77.352	77.352
01/08/2016-31/08/2016	11,792.880	11,792.880	11,792.880	78.936	78.936	78.936
01/09/2016-30/09/2016	14,611.080	14,611.080	14,611.080	101.904	101.904	101.904
01/10/2016-31/10/2016	15,267.120	15,267.120	15,267.120	50.424	50.424	50.424
01/11/2016-30/11/2016	21,792.940	21,792.940	21,792.940	44.088	44.088	44.088
01/12/2016-31/12/2016	21,667.270	21,667.270	21,667.270	20.592	20.592	20.592

Subtotal 2016	-	-	222,773.760	-	-	652.608
Total	-	-	1,187,369.330	-	-	3,764.640

Table 2: Electricity supplied by the Project B to the grid and Net electricity supplied to the grid by the project activity

Monitoring period	EG _{export,other,y}					EG _y
	Data based on meter readings			Data based on sales receipt	Conservative value	Calculated by EG _{export,total} - EG _{import,total} - EG _{export,other}
	Meter readings based on M7	Meter readings based on M8	Subtotal			
	MWh	MWh	MWh			
	G	H	I=G+H	J	K	L=C-F-K
26/10/2011-24/11/2011	6,205.850	4,519.340	10,725.190	10,725.190	10,725.190	9,407.290
25/11/2011-25/12/2011	5,183.500	3,701.180	8,884.680	8,884.680	8,884.680	12,033.680
26/12/2011-31/12/2011	940.450	626.500	1,566.950	1,566.950	1,566.950	1,838.650
Subtotal 2011	-	-	-	-	21,176.820	23,279.620
01/01/2012-25/01/2012	2,731.400	1,868.720	4,600.120	4,600.120	4,600.120	5,043.800
26/01/2012-24/02/2012	5,414.150	3,664.360	9,078.510	9,078.510	9,078.510	9,602.130
25/02/2012-25/03/2012	5,057.150	3,866.240	8,923.390	8,923.390	8,923.390	10,016.890
26/03/2012-24/04/2012	9,372.930	7,111.300	16,484.230	16,484.230	16,484.230	15,734.330
25/04/2012-25/05/2012	5,056.100	4,210.220	9,266.320	9,266.320	9,266.320	9,934.400
26/05/2012-30/06/2012	6,950.300	5,114.900	12,065.200	12,065.200	12,065.200	12,969.920
01/07/2012-31/07/2012	4,236.400	2,774.520	7,010.920	7,010.920	7,010.920	7,593.560
01/08/2012-23/08/2012	1,924.300	1,245.160	3,169.460	3,169.460	3,169.460	3,198.220
24/08/2012-30/09/2012	5,205.200	3,252.760	8,457.960	8,457.960	8,457.960	9,454.440
01/10/2012-31/10/2012	5,520.900	3,970.120	9,491.020	9,491.020	9,491.020	10,171.700
01/11/2012-30/11/2012	7,530.250	5,731.600	13,261.850	13,261.850	13,261.850	13,671.430
01/12/2012-31/12/2012	6,933.150	4,899.160	11,832.310	11,832.310	11,832.310	12,674.810
Subtotal 2012	-	-	-	-	113,641.290	120,065.630
01/01/2013-31/01/2013	5,216.750	3,450.720	8,667.470	8,667.470	8,667.470	9,598.690

01/02/2013-28/02/2013	3,555.650	2,603.160	6,158.810	6,158.810	6,158.810	6,301.990
01/03/2013-31/03/2013	8,317.400	7,288.680	15,606.080	15,606.080	15,606.080	16,034.320
01/04/2013-30/04/2013	8,352.750	6,558.160	14,910.910	14,910.910	14,910.910	15,161.330
01/05/2013-31/05/2013	6,647.900	4,994.080	11,641.980	11,641.980	11,641.980	11,941.140
01/06/2013-30/06/2013	5,439.350	3,862.880	9,302.230	9,302.230	9,302.230	10,070.090
01/07/2013-31/07/2013	3,877.300	2,582.160	6,459.460	6,459.460	6,459.460	6,912.140
01/08/2013-31/08/2013	4,000.150	2,795.240	6,795.390	6,795.390	6,795.390	7,471.170
01/09/2013-30/09/2013	3,384.150	2,388.680	5,772.830	5,772.830	5,772.830	6,120.370
01/10/2013-31/10/2013	5,784.450	4,340.840	10,125.290	10,125.290	10,125.290	10,619.830
01/11/2013-30/11/2013	6,249.250	4,678.240	10,927.490	10,927.490	10,927.490	11,686.750
01/12/2013-31/12/2013	6,807.150	4,745.440	11,552.590	11,552.590	11,552.590	12,621.890
Subtotal 2013	-	-	-	-	117,920.530	124,539.710
01/01/2014-31/01/2014	5,553.450	4,263.000	9,816.450	9,816.450	9,816.450	10,292.430
01/02/2014-28/02/2014	5,750.500	4,889.080	10,639.580	10,639.580	10,639.580	10,517.380
01/03/2014-31/03/2014	6,045.550	4,866.680	10,912.230	10,912.230	10,912.230	10,545.690
01/04/2014-30/04/2014	5,184.200	3,971.240	9,155.440	9,155.440	9,155.440	9,760.160
01/05/2014-31/05/2014	7,998.900	6,256.040	14,254.940	14,254.940	14,254.940	15,640.420
01/06/2014-30/06/2014	3,493.000	2,579.920	6,072.920	6,072.920	6,072.920	5,395.240
01/07/2014-31/07/2014	4,049.150	3,115.000	7,164.150	7,164.150	7,164.150	8,007.930
01/08/2014-31/08/2014	1,942.500	1,285.480	3,227.980	3,227.980	3,227.980	3,018.260
01/09/2014-30/09/2014	2,501.450	1,855.560	4,357.010	4,357.010	4,357.010	4,626.910
01/10/2014-31/10/2014	5,845.700	4,852.680	10,698.380	10,698.380	10,698.380	11,546.260
01/11/2014-30/11/2014	4,472.650	3,355.240	7,827.890	7,827.890	7,827.890	7,901.230
01/12/2014-31/12/2014	7,835.450	6,312.040	14,147.490	14,147.490	14,147.490	15,288.510
Subtotal 2014	-	-	-	-	108,274.460	112,540.420

01/01/2015-31/01/2015	6,725.600	5,237.400	11,963.000	11,963.000	11,963.000	11,828.680
01/02/2015-28/02/2015	4,585.000	3,462.480	8,047.480	8,047.480	8,047.480	7,993.160
01/03/2015-31/03/2015	6,470.100	5,278.280	11,748.380	11,748.380	11,748.380	13,339.540
01/04/2015-30/04/2015	8,271.200	6,502.720	14,773.920	14,773.920	14,773.920	13,697.950
01/05/2015-31/05/2015	6,485.150	4,981.200	11,466.350	11,466.350	11,466.350	11,447.530
01/06/2015-30/06/2015	4,404.750	3,518.480	7,923.230	7,923.230	7,923.230	8,514.206
01/07/2015-31/07/2015	3,289.650	2,311.120	5,600.770	5,600.770	5,600.770	5,363.414
01/08/2015-31/08/2015	1,612.800	1,270.640	2,883.440	2,883.440	2,883.440	2,729.730
01/09/2015-30/09/2015	2,129.750	1,258.880	3,388.630	3,388.630	3,388.630	4,136.954
01/10/2015-31/10/2015	5,756.800	4,335.520	10,092.320	10,092.320	10,092.320	8,835.422
01/11/2015-30/11/2015	6,378.050	4,868.920	11,246.970	11,246.970	11,246.970	11,582.434
01/12/2015-31/12/2015	5,950.350	4,525.080	10,475.430	10,475.430	10,475.430	10,966.118
Subtotal 2015	-	-	-	-	109,609.920	110,435.138
01/01/2016-31/01/2016	6,329.750	4,700.080	11,029.830	11,029.830	11,029.830	11,502.046
01/02/2016-28/02/2016	4,445.350	3,394.720	7,840.070	7,840.070	7,840.070	8,040.060
01/03/2016-31/03/2016	6,266.050	4,950.120	11,216.170	11,216.170	11,216.170	11,072.030
01/04/2016-30/04/2016	6,717.900	5,052.880	11,770.780	11,770.780	11,770.780	12,252.160
01/05/2016-31/05/2016	5,243.000	3,844.400	9,087.400	9,087.400	9,087.400	9,165.824
01/06/2016-30/06/2016	5,472.600	4,028.920	9,501.520	9,501.520	9,501.520	9,898.518
01/07/2016-31/07/2016	3,997.000	2,877.560	6,874.560	6,874.560	6,874.560	8,034.838
01/08/2016-31/08/2016	3,522.400	2,393.160	5,915.560	5,915.560	5,915.560	5,798.384
01/09/2016-30/09/2016	3,993.150	3,061.800	7,054.950	7,054.950	7,054.950	7,454.226
01/10/2016-31/10/2016	4,307.800	3,166.240	7,474.040	7,474.040	7,474.040	7,742.656
01/11/2016-30/11/2016	6,022.100	4,545.800	10,567.900	10,567.900	10,567.900	11,180.952
01/12/2016-31/12/2016	5,990.600	4,446.400	10,437.000	10,437.000	10,437.000	11,209.678

Subtotal 2016	-	-	-	-	108,769.780	113,351.372
Total	-	-	-	-	579,392.800	604,211.890

Table 3: Electricity supplied by the project activity to the grid, Electricity imported from the grid by the project and Net electricity supplied to the grid by the project activity

Monitoring period	EG _{export,project}				EG _{import,project}				EG _y
	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Calculated by EG _{export,project} - EG _{import,project}
	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh
	M	N	O	P=M+N+O	Q	R	S	P=M+N+O	U=P-T
26/10/2011-24/11/2011	2,915.052	4,772.488	2,261.560	9,949.100	9.912	16.184	8.092	34.188	9,914.912
25/11/2011-25/12/2011	3,335.332	5,821.144	2,158.212	11,314.688	8.428	13.300	7.028	28.756	11,285.932
26/12/2011-31/12/2011	632.016	1,033.844	506.744	2,172.604	1.260	1.988	1.064	4.312	2,168.292
Subtotal 2011	-	-	-	23,436.392	-	-	-	67.256	23,369.136
01/01/2012-25/01/2012	1,454.348	2,373.140	1,174.936	5,002.424	10.528	19.096	8.988	38.612	4,963.812
26/01/2012-24/02/2012	2,835.392	4,916.044	2,305.772	10,057.208	9.968	12.404	8.092	30.464	10,026.744
25/02/2012-25/03/2012	2,610.832	4,486.692	2,319.828	9,417.352	5.628	8.456	4.340	18.424	9,398.928
26/03/2012-24/04/2012	4,720.996	8,334.396	4,169.536	17,224.928	5.180	7.056	3.444	15.680	17,209.248
25/04/2012-25/05/2012	2,762.452	4,904.424	2,473.156	10,140.032	10.808	14.868	7.448	33.124	10,106.908
26/05/2012-30/06/2012	3,483.256	6,663.888	3034.948	13,182.092	8.932	13.776	6.860	29.568	13,152.524
01/07/2012-31/07/2012	1,604.456	3,922.548	1,711.220	7,238.224	9.744	18.200	9.716	37.660	7,200.564
01/08/2012-23/08/2012	914.900	1,664.264	761.180	3,340.344	11.508	17.052	9.212	37.772	3,302.572
24/08/2012-30/09/2012	2,820.160	4,776.576	2,132.872	9,729.608	11.312	17.836	9.520	38.668	9,690.940
01/10/2012-31/10/2012	2,871.484	5,091.212	2,436.280	10,398.976	6.216	9.800	4.144	20.160	10,378.816
01/11/2012-30/11/2012	3,851.792	6,986.056	3,082.212	13,920.060	4.956	7.280	3.752	15.988	13,904.072
01/12/2012-31/12/2012	3,763.564	6,190.408	2,989.308	12,943.280	5.432	9.016	4.760	19.208	12,924.072
Subtotal 2012	-	-	-	122,594.528	-	-	-	335.328	122,259.200

01/01/2013-31/01/2013	2,759.120	4,700.528	2,327.556	9,787.204	4.032	8.036	4.088	16.156	9,771.048
01/02/2013-28/02/2013	1,784.916	3,222.492	1,579.816	6,587.224	12.992	21.056	10.472	44.520	6,542.704
01/03/2013-31/03/2013	2,943.528	7,795.256	4,116.616	14,855.400	2.576	6.188	2.996	11.760	14,843.640
01/04/2013-30/04/2013	3,532.844	7,423.892	3,783.612	14,740.348	3.052	5.376	2.772	11.200	14,729.148
01/05/2013-31/05/2013	3,238.312	5,902.764	3,025.456	12,166.532	5.796	7.812	3.976	17.584	12,148.948
01/06/2013-30/06/2013	2,748.620	5,187.112	2,387.112	10,322.844	9.576	13.888	5.936	29.400	10,293.444
01/07/2013-31/07/2013	1,912.764	3,592.708	1,645.420	7,150.892	14.672	22.316	11.424	48.412	7,102.480
01/08/2013-31/08/2013	2,085.944	3,855.292	1,754.060	7,695.296	10.808	16.408	7.532	34.748	7,660.548
01/09/2013-30/09/2013	1,716.904	3,042.424	1,474.564	6,233.892	12.572	18.144	9.884	40.600	6,193.292
01/10/2013-31/10/2013	3,113.264	5,326.804	2,546.572	10,986.640	6.776	9.296	4.648	20.720	10,965.920
01/11/2013-30/11/2013	3,316.236	5,699.820	2,778.076	11,794.132	6.104	9.156	4.592	19.852	11,774.280
01/12/2013-31/12/2013	3,745.028	6,252.344	2,975.420	12,972.792	3.108	4.452	2.184	9.744	12,963.048
Subtotal 2013	-	-	-	125,293.196	-	-	-	304.696	124,988.500
01/01/2014-31/01/2014	2,925.580	5,157.600	2,649.780	10,732.960	3.920	5.432	2.884	12.236	10,720.724
01/02/2014-28/02/2014	2,846.396	5,099.780	2,639.728	10,585.904	6.216	9.408	5.208	20.832	10,565.072
01/03/2014-31/03/2014	2,911.776	5,364.660	2,800.812	11,077.248	6.636	9.996	5.068	21.700	11,055.548
01/04/2014-30/04/2014	2,656.528	4,629.464	2,401.448	9,687.440	8.764	12.320	6.244	27.328	9,660.112
01/05/2014-31/05/2014	4,172.728	7,564.284	3,759.868	15,496.880	4.704	6.328	2.884	13.916	15,482.964
01/06/2014-30/06/2014	1,581.412	2,890.216	1,458.688	5,930.316	11.452	17.528	8.708	37.688	5,892.628
01/07/2014-31/07/2014	2,225.832	4,095.644	1,916.124	8,237.600	10.892	16.240	8.484	35.616	8,201.984
01/08/2014-31/08/2014	926.212	1,643.236	801.444	3,370.892	15.708	23.940	11.984	51.632	3,319.260
01/09/2014-30/09/2014	1,556.352	2,581.628	1,349.096	5,487.076	8.876	13.972	7.588	30.436	5,456.640
01/10/2014-31/10/2014	3,077.284	5,411.224	2,741.368	11,229.876	2.000	12.712	6.524	21.236	11,208.640
01/11/2014-30/11/2014	2,455.600	3,929.968	2,074.268	8,459.836	6.636	9.548	5.292	21.476	8,438.360
01/12/2014-31/12/2014	4,517.408	6,672.120	4,027.156	15,216.684	3.808	5.516	2.688	12.012	15,204.672

Subtotal 2014	-	-	-	115,512.712	-	-	-	306.108	115,206.604
01/01/2015-31/01/2015	3,268.692	5,441.156	2,891.616	11,601.464	4.256	6.132	3.052	13.440	11,588.024
01/02/2015-28/02/2015	2,271.724	4,050.508	2,087.876	8,410.108	7.392	10.696	6.244	24.332	8,385.776
01/03/2015-31/03/2015	3,654.952	6,568.436	3,367.140	13,590.528	5.068	7.000	3.388	15.456	13,575.072
01/04/2015-30/04/2015	3,857.700	6,796.384	3,509.296	14,163.380	6.832	8.932	4.620	20.384	14,142.996
01/05/2015-31/05/2015	3,377.304	5,617.836	3,091.088	12,086.228	7.924	9.772	5.292	22.988	12,063.240
01/06/2015-30/06/2015	2,234.288	3,919.692	2,036.580	8,190.560	11.060	15.176	8.708	34.944	8,155.616
01/07/2015-31/07/2015	1,615.740	2,673.244	1,386.308	5,675.292	11.088	15.736	8.568	35.392	5,639.900
01/08/2015-31/08/2015	868.756	1,425.060	771.232	3,065.048	14.784	19.936	11.564	46.284	3,018.764
01/09/2015-30/09/2015	1,206.072	1,870.484	1,018.052	4,094.608	17.360	23.968	14.140	55.468	4,039.140
01/10/2015-31/10/2015	2,865.016	4,237.016	2,564.156	9,666.188	6.300	8.792	4.956	20.048	9,646.140
01/11/2015-30/11/2015	3,575.684	5,464.872	2,944.564	11,985.120	5.600	8.456	4.620	18.676	11,966.444
01/12/2015-31/12/2015	2,962.372	5,585.608	2,753.380	11,301.360	8.008	11.928	6.412	26.348	11,275.012
Subtotal 2015	-	-	-	113,829.884	-	-	-	333.760	113,496.124
01/01/2016-31/01/2016	3,247.328	5,635.280	2,857.596	11,740.204	4.760	7.476	3.640	15.876	11,724.328
01/02/2016-28/02/2016	2,314.368	4,303.292	2,056.348	8,674.008	7.168	10.556	5.740	23.464	8,650.544
01/03/2016-31/03/2016	3,114.496	5,513.648	2,957.948	11,586.092	3.920	4.424	2.100	10.444	11,575.648
01/04/2016-30/04/2016	3,316.124	5,883.752	2,992.248	12,192.124	4.704	6.412	3.080	14.196	12,177.928
01/05/2016-31/05/2016	2,545.088	4,680.676	2,366.700	9,592.464	6.664	9.016	4.256	19.936	9,572.528
01/06/2016-30/06/2016	2,738.988	4,991.028	2,443.588	10,173.604	7.840	11.004	5.292	24.136	10,149.468
01/07/2016-31/07/2016	2,036.076	3,676.260	1,656.788	7,369.124	12.068	18.340	9.296	39.704	7,329.420
01/08/2016-31/08/2016	1,763.328	3,041.640	1,475.124	6,280.092	10.976	15.540	7.756	34.272	6,245.820
01/09/2016-30/09/2016	2,012.696	3,810.044	1,907.528	7,730.268	12.432	19.180	9.912	41.524	7,688.744
01/10/2016-31/10/2016	2,238.208	3,757.852	1,972.544	7,968.604	6.076	8.428	5.040	19.544	7,949.060
01/11/2016-30/11/2016	3,158.876	5,493.124	2,744.084	11,396.084	4.648	6.244	3.472	14.364	11,381.720

01/12/2016-31/12/2016	3,287.172	5,370.232	2,736.076	11,393.480	5.936	8.876	4.452	19.264	11,374.216
Subtotal 2016	-	-	-	116,096.148	-	-	-	276.724	115,819.424
Total	-	-	-	616,762.860	-	-	-	1,623.872	615,138.988

Table 4: Net electricity supplied to the grid by the project activity

Monitoring period	EG _y		
	Calculated by $EG_{\text{export, total}} - EG_{\text{import, total}} - EG_{\text{export, other, y}}$	Calculated by $EG_{\text{export, project}} - EG_{\text{import, project}}$	Conservative value
	Unit: MWh	Unit: MWh	Unit: (tCO ₂ e)
	L	U	W=Min(L,U)
26/10/2011-24/11/2011	9,407.290	9,914.912	9,407.290
25/11/2011-25/12/2011	12,033.680	11,285.932	11,285.932
26/12/2011-31/12/2011	1,838.650	2,168.292	1,838.650
Subtotal 2011	23,279.620	23,369.136	22,531.872
01/01/2012-25/01/2012	5,043.800	4,963.812	4,963.812
26/01/2012-24/02/2012	9,602.130	10,026.744	9,602.130
25/02/2012-25/03/2012	10,016.890	9,398.928	9,398.928
26/03/2012-24/04/2012	15,734.330	17,209.248	15,734.330
25/04/2012-25/05/2012	9,934.400	10,106.908	9,934.400
26/05/2012-30/06/2012	12,969.920	13,152.524	12,969.920
01/07/2012-31/07/2012	7,593.560	7,200.564	7,200.564
01/08/2012-23/08/2012	3,198.220	3,302.572	3,198.220
24/08/2012-30/09/2012	9,454.440	9,690.940	9,454.440
01/10/2012-31/10/2012	10,171.700	10,378.816	10,171.700
01/11/2012-30/11/2012	13,671.430	13,904.072	13,671.430
01/12/2012-31/12/2012	12,674.810	12,924.072	12,674.810
Subtotal 2012	120,065.630	122,259.200	118,974.684
01/01/2013-31/01/2013	9,598.690	9,771.048	9,598.690
01/02/2013-28/02/2013	6,301.990	6,542.704	6,301.990

01/03/2013-31/03/2013	16,034.320	14,843.640	14,843.640
01/04/2013-30/04/2013	15,161.330	14,729.148	14,729.148
01/05/2013-31/05/2013	11,941.140	12,148.948	11,941.140
01/06/2013-30/06/2013	10,070.090	10,293.444	10,070.090
01/07/2013-31/07/2013	6,912.140	7,102.480	6,912.140
01/08/2013-31/08/2013	7,471.170	7,660.548	7,471.170
01/09/2013-30/09/2013	6,120.370	6,193.292	6,120.370
01/10/2013-31/10/2013	10,619.830	10,965.920	10,619.830
01/11/2013-30/11/2013	11,686.750	11,774.280	11,686.750
01/12/2013-31/12/2013	12,621.890	12,963.048	12,621.890
Subtotal 2013	124,539.710	124,988.500	122,916.848
01/01/2014-31/01/2014	10,292.430	10720.724	10,292.430
01/02/2014-28/02/2014	10,517.380	10565.072	10,517.380
01/03/2014-31/03/2014	10,545.690	11055.548	10,545.690
01/04/2014-30/04/2014	9,760.160	9660.112	9,660.112
01/05/2014-31/05/2014	15,640.420	15482.964	15,482.964
01/06/2014-30/06/2014	5,395.240	5892.628	5,395.240
01/07/2014-31/07/2014	8,007.930	8201.984	8,007.930
01/08/2014-31/08/2014	3,018.260	3319.260	3,018.260
01/09/2014-30/09/2014	4,626.910	5456.640	4,626.910
01/10/2014-31/10/2014	11,546.260	11,208.640	11,208.640
01/11/2014-30/11/2014	7,901.230	8,438.360	7,901.230
01/12/2014-31/12/2014	15,288.510	15204.672	15,204.672
Subtotal 2014	112,540.420	115,206.604	111,861.458
01/01/2015-31/01/2015	11,828.680	11,588.024	11,588.024

01/02/2015-28/02/2015	7,993.160	8,385.776	7,993.160
01/03/2015-31/03/2015	13,339.540	13,575.072	13,339.540
01/04/2015-30/04/2015	13,697.950	14,142.996	13,697.950
01/05/2015-31/05/2015	11,447.530	12,063.240	11,447.530
01/06/2015-30/06/2015	8,514.206	8,155.616	8,155.616
01/07/2015-31/07/2015	5,363.414	5,639.900	5,363.414
01/08/2015-31/08/2015	2,729.730	3,018.764	2,729.730
01/09/2015-30/09/2015	4,136.954	4,039.140	4,039.140
01/10/2015-31/10/2015	8,835.422	9,646.140	8,835.422
01/11/2015-30/11/2015	11,582.434	11,966.444	11,582.434
01/12/2015-31/12/2015	10,966.118	11,275.012	10,966.118
Subtotal 2015	110,435.138	113,496.124	109,738.078
01/01/2016-31/01/2016	11,502.046	11,724.328	11,502.046
01/02/2016-28/02/2016	8,040.060	8,650.544	8,040.060
01/03/2016-31/03/2016	11,072.030	11,575.648	11,072.030
01/04/2016-30/04/2016	12,252.160	12,177.928	12,177.928
01/05/2016-31/05/2016	9,165.824	9,572.528	9,165.824
01/06/2016-30/06/2016	9,898.518	10,149.468	9,898.518
01/07/2016-31/07/2016	7,407.838	7,329.420	7,329.420
01/08/2016-31/08/2016	5,798.384	6,245.820	5,798.384
01/09/2016-30/09/2016	7,454.226	7,688.744	7,454.226
01/10/2016-31/10/2016	7,742.656	7,949.060	7,742.656
01/11/2016-30/11/2016	11,180.952	11,381.720	11,180.952
01/12/2016-31/12/2016	11,209.678	11,374.216	11,209.678
Subtotal 2016	113,351.372	115,819.424	112,571.722

Total	604,211.890	615,138.988	598,594.662
--------------	--------------------	--------------------	--------------------

The corresponding baseline emission reductions during this monitoring period is calculated as:

Table 5: Calculation of baseline emission reductions

Monitoring period	EG _y	EF _{grid,CM,y}	BE _y
	Unit: MWh	Unit: tCO ₂ e/MWh	Unit: tCO ₂
26/10/2011-31/12/2011	22,531.872	1.0548	23,766
01/01/2012-31/12/2012	118,974.684	1.0548	125,494
01/01/2013-31/12/2013	122,916.848	1.0548	129,652
01/01/2014-31/12/2014	111,861.458	1.0548	117,991
01/01/2015-31/12/2015	109,738.078	1.0548	115,751
01/01/2016-31/12/2016	112,571.722	1.0548	118,740
Total	598,594.662	1.0548	631,397

Project emissions

According to ACM0002(version 07), the project does not have any GHG emissions.

$$PE_y=0$$

Leakages

According to ACM0002 (Version 07), the leakage from the Project is zero.

$$L_y=0$$

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals(tCO ₂ e)
26/10/2011-31/12/2011	23,766	0	0	23,766
01/01/2012-31/12/2012	125,494	0	0	125,494
01/01/2013-31/12/2013	129,652	0	0	129,652
01/01/2014-31/12/2014	117,991	0	0	117,991
01/01/2015-31/12/2015	115,751	0	0	115,751
01/01/2016-31/12/2016	118,740	0	0	118,740
Total	631,397	0	0	631,397

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered CDM-PDD

As per the registered PDD, the estimated quantity of annual emission reductions for the project activity is 107,488 tCO₂e. The estimated emission reductions during this monitoring period from 26/10/2011 to 31/12/2016 (1893 days) are 557,465.162 tCO₂e while the actual emission reductions are 631,397 tCO₂e, which is 13% higher than the estimation emission reductions.

This would not affect the additionality of the project. As per to the registered PDD, the IRR is still less than the beach mark of 8% when the net electricity supplied to the grid increase 13%. Only when the electricity generation increase by 16%, the IRR could reach benchmark.

The reasons for electricity and emission reductions increase are as follows: The data source of annual electricity supplied to the grid from the project in the registered PDD is from the Feasibility Study Report (FSR) of the project, which is calculated based on the monitored wind resources data from the project onsite anemometer tower during year 2005 to 2006 and the monitored wind resources data from local Weather Station during year 1986 to 2006.

Comparing the yearly average wind from 2012-2016 and that from 1987- 2006, the wind speed in this period is 65% higher than the data used in the FSR (refer to figure 1 for details). Furthermore, the Opti Speed generators have been used by the project, which allows the rotor speed to vary within a range of approximately 60 percent in relation to normal rpm. So it's reasonable that the net electricity supplied to the grid by the project during this monitoring period is 13% higher than the estimated data because of the rich wind resources, Opti Speed generators and the better management.

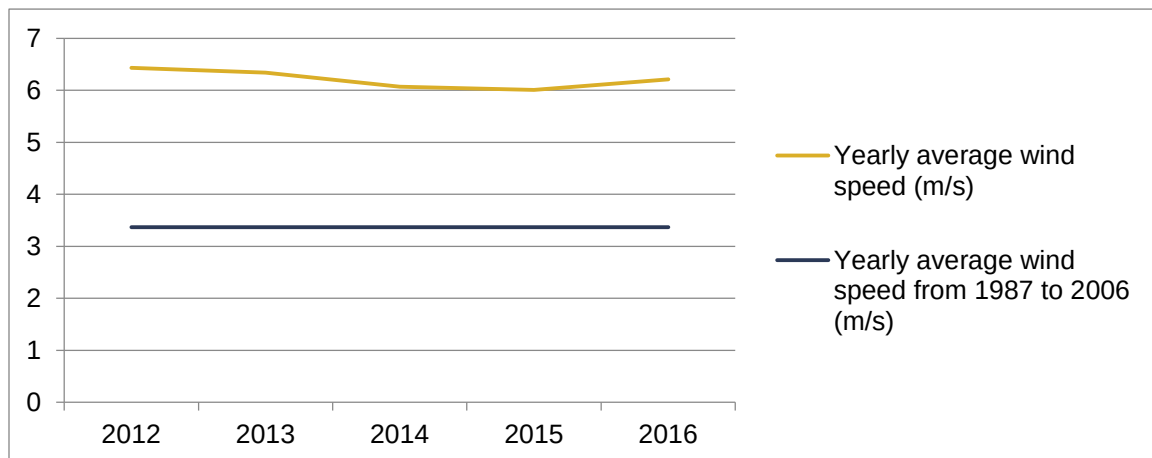


Figure 1. Comparison of wind speed from 2012-2016 with 1987-2006

CL01 was raised because the estimated emission reductions during this monitoring period from 26/10/2011 to 31/12/2016 (1893 days) are 557,465.162 tCO₂e while the actual emission reductions are 631,397 tCO₂e, which is 13% higher than the estimation emission reductions. Further clarification is sought on the differences of emission reduction. The differences in emissions reductions have been clarified in section 5.4 of the revised MR, thus **CL 01** was subsequently successfully closed. Refer to Appendix B for details.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CQC concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

China Quality Certification Centre (CQC) has performed the verification of the emission reductions that have been reported for the project activity “Shandong Penglai Pingdingshan Wind Farm Project” in China for the period 26/10/2011-31/12/2016.

The verification is based on the baseline and monitoring methodology ACM0002 (version 6.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable
- and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and

- is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CQC's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CQC planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CQC does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CQC and the engagement conditions detailed in this report. CQC cannot be held liable by any party for decisions made or not made based on this report.

In CQC's opinion the GHG emissions reductions of the "Shandong Penglai Pingdingshan Wind Farm Project" for the period 26/10/2011-31/12/2016 are fairly stated in the monitoring report (version 2.0 dated 28/08/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 07) and the monitoring plan contained in the registered VCS PD.

CQC can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CQC confirms the following statement:

Reporting period: 26/10/2011-31/12/2016

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
26/10/2011-31/12/2011	23,766	0	0	23,766
01/01/2012-31/12/2012	125,494	0	0	125,494
01/01/2013-31/12/2013	129,652	0	0	129,652
01/01/2014-31/12/2014	117,991	0	0	117,991
01/01/2015-31/12/2015	115,751	0	0	115,751
01/01/2016-31/12/2016	118,740	0	0	118,740
Total	631,397	0	0	631,397

APPENDIX A: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.: VER/VCU Monitoring Report for Shandong Penglai Pingdingshan Wind Farm Project, V01 dated 29/07/2022, version 02 dated 28/08/2022
/2/	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.: Emission reduction calculation spreadsheet for Shandong Penglai Pingdingshan Wind Farm Project, version 1.0 dated 29/07/2022 and version 2.0 dated 05/09/2022
/3/	China Resources Wind Power (Yantai) Co., Ltd. and State Grid Shandong Electric Power Company Yantai Power Supply Company : Power Purchase Agreement for Shandong Penglai Pingdingshan Wind Farm Project
/4/	China Resources Wind Power (Yantai) Co., Ltd.: Diagram of power connection system
/5/	China Resources Wind Power (Yantai) Co., Ltd.: VER monitoring manual and management procedure.
/6/	China Resources Wind Power (Yantai) Co., Ltd.: Records of training for on-site staff.
/7/	China Resources Wind Power (Yantai) Co., Ltd.: Daily operation and maintenance records
/8/	China Resources Wind Power (Yantai) Co., Ltd.: Monthly and daily reading records of Shandong Penglai Pingdingshan Wind Farm Project from 26/10/2011-31/12/2016
/9/	State Grid Shandong Electric Power Company.: Monthly electricity receipts from 26/10/2011-31/12/2016 & Confirmation letter for electricity readings the period 26/10/2011-31/12/2016 issued by the grid company
/10/	Comment records
/11/	Contact information of PP
/12/	Yearly questionnaires for local stakeholders
/13/	China Resources Wind Power (Yantai) Co., Ltd.: Registered CDM PDD (version 05 dated 20/07/2009)
/14/	State Grid Shandong Electric Power Company: Calibration certificates for meters M1,M2, M4, M5, M6,M7,M8,M9 and M10 covering this monitoring period Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and

	Environment of China
/15/	Business license of China Resources Wind Power (Yantai) Co., Ltd.
/16/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2000)
/17/	UNFCCC, http://cdm.unfccc.int
/18/	Gold Standard, http://www.goldstandard.org/
/19/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/20/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/21/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/22/	China Resources Wind Power (Yantai) Co., Ltd.: Staff roster in 2016
/23/	China Resources Wind Power (Yantai) Co., Ltd.: the Explanation of the employment offered by the project.
/24/	TÜV NORD CERT GmbH JI/CDM Certification Program: VCS Validation Report, version 05, dated 20/07/2009
/25/	VER/VCU Monitoring Report for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 20/12/2008-20/07/2009
/26/	Bureau Veritas Certification Holding SAS: VCS Verification Report for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 20/12/2008-20/07/2009
/27/	CDM/CER Monitoring Reports for Shandong Penglai Pingdingshan Wind Farm Project of previous CDM monitoring periods 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011~25/10/2011
/28/	CDM Verification Reports for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011~25/10/2011
/29/	Nameplates of the equipment of the project
/30/	Stakeholder Consultation Opinion Response policy of Shandong Penglai Pingdingshan Wind Farm Project
/31/	Feasibility Study Report of Shandong Penglai Pingdingshan Wind Farm Project

/32/	Calibration records of State Grid Shandong Electric Power Company Yantai Power Supply Company
------	---

Methodologies, tools and other guidance

/33/	Verified Carbon Standard: VCS Standard, version 4.3
/34/	Verified Carbon Standard: VCS Program Guide, version 4.1
/35/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/36/	Verified Carbon Standard: Registration and Issuance Process, version 4.1
/37/	UNFCCC EB: Approved methodology, ACM0002 , version 07

Persons interviewed

/38/	Mr. Gu Haiguang, Duty Director, China Resources Wind Power (Yantai) Co., Ltd. Mr. ZhouWei, O&M Staff, China Resources Wind Power (Yantai) Co., Ltd. Mr. Zhao Jie, Carbon asset manager, China Resources Wind Power (Yantai) Co., Ltd. Ms. Li Lin, Project Manager, Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd. Ms. Ding Xue, Technical director, Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.
------	---

APPENDIX B: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
CAR 01	The monitoring period in the MR (from 25/06/2012 to 19/12/2018, which is more than six consecutive years) does not meet the requirements of rotation of validation/verification bodies in VCS-Standard_v4.3.	The monitoring period has revised from 25/06/2012-19/12/2018 to 26/10/2011-31/12/2016. As described in section 1.9 of MR V1.0, CERs during the 4th monitoring period from 26/10/2011 to 24/06/2012 have not been issued and the verification contract between PP and VVB has been terminated. On 15/08/2022, the monitoring report withdrawal request form has been submitted to EB though the DOE BVQI.	OK. It has been revised and verified by CQC by checking updated MR. Therefore, CAR 01 is closed.
CAR 02	The information of the	The information of the	OK.

	monitoring meters installed during the monitoring period is not provided in the MR and supporting documents.	monitoring meters installed during the monitoring period has been added in Section 4.2 of the revised MR, and the supporting document has been replenished.	It has been revised and verified by CQC by checking updated MR. Therefore, CAR 02 is closed.
--	--	---	--

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	The estimated emission reductions during this monitoring period from 26/10/2011 to 31/12/2016 (1893 days) are 557,465.162 tCO₂e while the actual emission reductions are 631,397 tCO₂e, which is 13% higher than the estimation emission reductions. Further clarification is sought on the differences of emission reduction.	According to the wind resource monitoring data from the wind power farm, the average wind speed in the project site during 2012-2016 was 6.21 m/s, which was nearly 65% higher than the average wind speed of 3.77 m/s during 1987-2006 which is basic data to prepare the Feasibility Study Report (FSR) of the project. The increase rate of power generation does not exceed the sensitivity threshold, which does not affect the additionality of the project. The differences in emissions reductions have been clarified in	OK. It has been revised and verified by CQC by checking updated MR. Therefore, CL 01 is closed.

		section 5.4 of the revised MR.	
--	--	--------------------------------	--

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA